Appendix/Technical Reference

This appendix is the **engineering / maintenance** reference. It intentionally contains the deeper technical details that are not placed in the client-facing docs. If you are an operator/client and you are only trying to use the system, you usually do not need this appendix.

A1. Bill of Materials

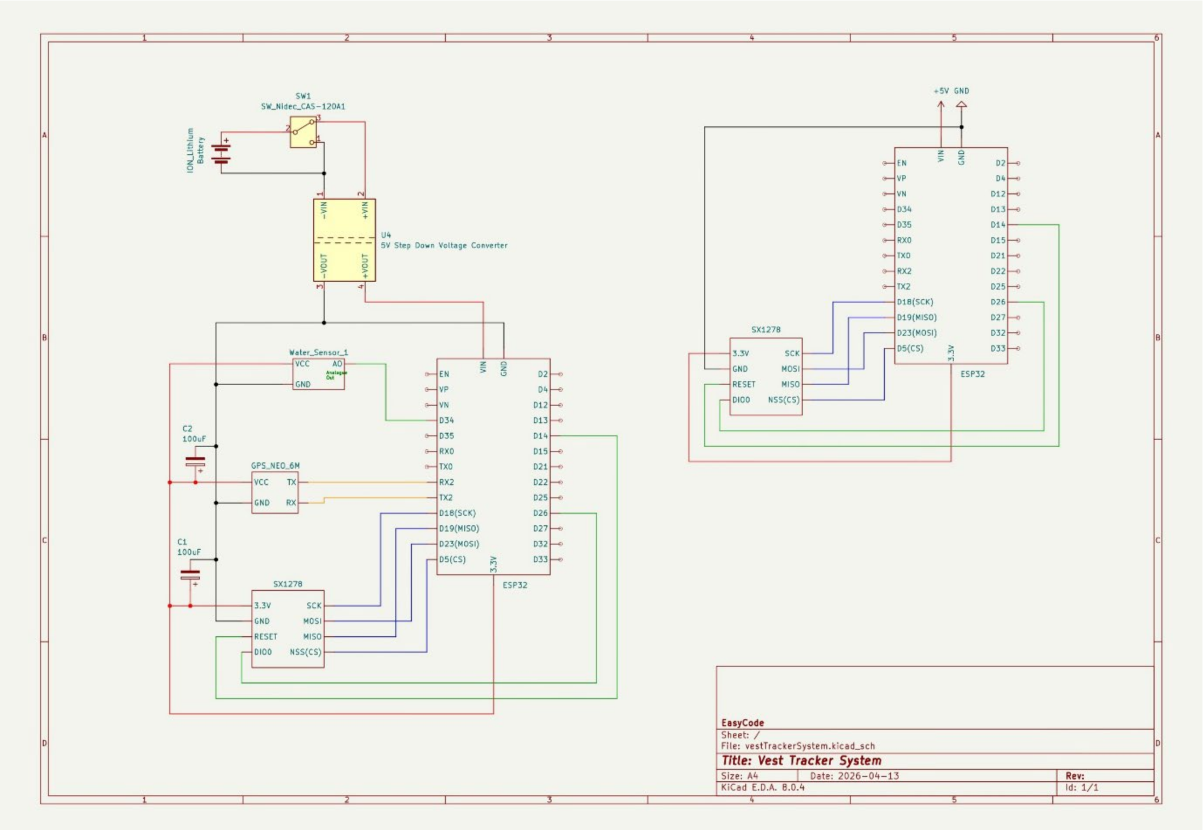
Vest Unit (Transmitter)

Component	Quantity	Notes
ESP32 DevKit V1	1	Main controller
LoRa SX1278 (Ra-02)	1	Wireless transmission
GPS Module GY-NEO6MV2	1	Location tracking
Rain Sensor (plate + module)	1 set	Water detection
Lithium Battery (18650)	3	Power source
Battery Holder (3-slot)	1	Holds batteries
DC-DC Step Down Regulator	2	One for 5V rail, one for 3.3V rail
Switch	1	Power control
Capacitor 100µF	1	For LoRa stability
Capacitor 10µF	1	For GPS stability
Wires	---	Connections
Perfboard	1	Mounting
Connectors	---	Connections

Control Tower (Receiver)

Component	Quantity	Notes
ESP32 DevKit V1	1	Main controller
LoRa SX1278 (Ra-02)	1	Receive data
Power Supply (USB)	1	Stable power
Jumper Wires	---	Connections
Perfboard	1	Mounting

A2. Circuit Schematic



A3. Pin Connections / Wiring Tables

Vest Unit Wiring (Power Chain)

Component	Pin/Label	Connected To	Notes
Battery Holder (3x Lithium)	+	Switch input	Main power
Battery Holder	–	Common GND	Ground
Switch	Output	DC-DC (5V rail) IN+ and DC-DC (3.3V rail) IN+	Power control
DC-DC (5V rail)	IN+	Switch output	Input power
DC-DC (5V rail)	IN–	GND	Common ground
DC-DC (5V rail)	OUT+ (5V)	ESP32 VIN	5V rail

Component	Pin/Label	Connected To	Notes
DC-DC (5V rail)	OUT–	GND	Common ground
DC-DC (3.3V rail)	IN+	Switch output	Input power
DC-DC (3.3V rail)	IN–	GND	Common ground
DC-DC (3.3V rail)	OUT+ (3.3V)	LoRa VCC, GPS VCC and rain sensor VCC	3.3V rail
DC-DC (3.3V rail)	OUT–	GND	Common ground

Vest Unit Wiring (ESP32 Pin Map)

ESP32	Connected To	Notes
VIN	5V (from regulator)	Power input
GND	Common GND	Shared ground
3V3	Not used for powering modules	Use dedicated DC-DC 3.3V rail instead
GPIO16 (RX2)	GPS TX	UART
GPIO17 (TX2)	GPS RX (optional)	Can leave unconnected
GPIO18	LoRa SCK	SPI
GPIO19	LoRa MISO	SPI
GPIO23	LoRa MOSI	SPI
GPIO5	LoRa NSS (CS)	SPI select
GPIO14	LoRa RST	Reset
GPIO26	LoRa DIO0	Interrupt
GPIO34	Rain Sensor AO	Analog input

LoRa (SX1278 / Ra-02) Wiring

LoRa Pin	Connected To	Notes
VCC	DC-DC (3.3V rail) OUT+	3.3V only (do not power from ESP32 3V3)
GND	GND	

LoRa Pin	Connected To	Notes
SCK	GPIO18	
MISO	GPIO19	
MOSI	GPIO23	
NSS	GPIO5	
RST	GPIO14	
DIO0	GPIO26	

GPS (GY-NEO6MV2) Wiring

GPS Pin	Connected To	Notes
VCC	DC-DC (3.3V rail) OUT+	3.3V only (do not power from ESP32 3V3)
GND	GND	
TX	GPIO16	Data to ESP32
RX	GPIO17 (optional)	Can leave open

Rain Sensor Module Wiring

Rain Module	Connected To	Notes
VCC	5V rail or 3.3V rail	Prefer 3.3V rail for stable
GND	GND	
AO	GPIO34	Analog reading
DO	Not used	Optional only

A4. Capacitor Placement (Reference)

Location	Capacitor	Connection
LoRa	100µF	3.3V rail to GND (near module)
GPS	10µF	3.3V rail to GND (near module)

A5. Assembly Notes (Reference Only)

Vest Unit Assembly (High level)

1. Wire the power chain: Battery holder to Switch to DC-DC (5V rail) and DC-DC (3.3V rail).
2. Connect GPS to ESP32 UART2.
3. Connect LoRa to ESP32 SPI (power LoRa from the DC-DC 3.3V rail, not ESP32 3V3).
4. Connect Rain Sensor analog output to the ESP32 analog input.
5. Add the recommended capacitors close to the LoRa and GPS modules.

Control Tower Assembly (High level)

1. Connect LoRa module to ESP32 (same pin map as vest LoRa wiring).
2. Power ESP32 via USB or regulated 5V supply.

A6. Optional Operator Quick Reference

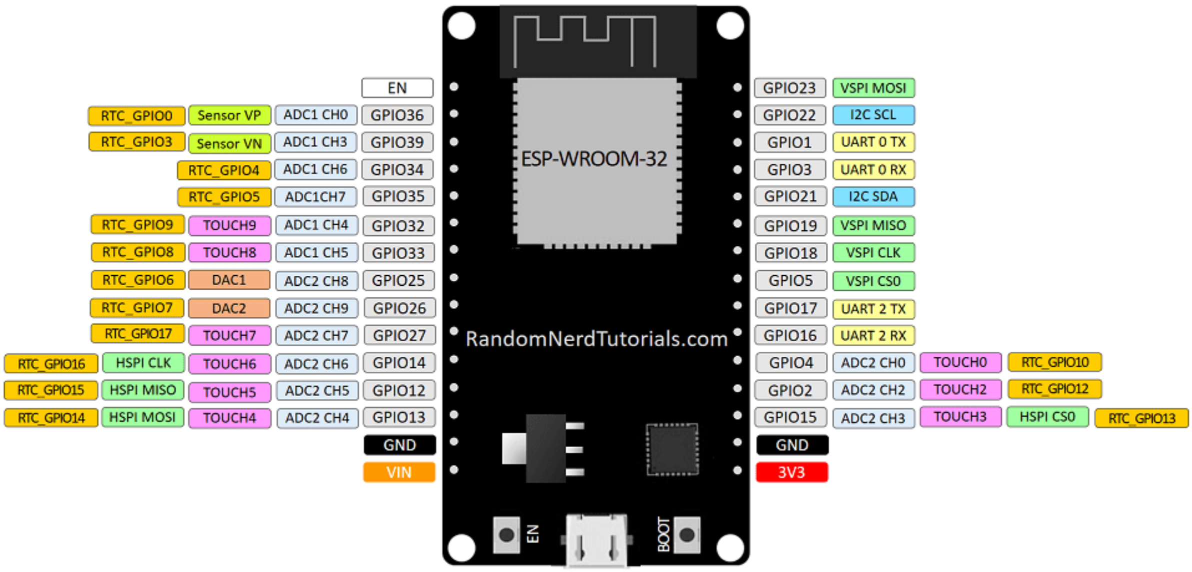
Glossary

1. **LoRa**: Long-range, low-power radio link used between the vest and control tower.
2. **Control tower**: The receiver unit that listens for LoRa packets and uploads them to the database.
3. **RTDB**: Firebase Realtime Database where control tower uploads packets when available.
4. **GPS lock**: GPS has a fresh valid position fix.

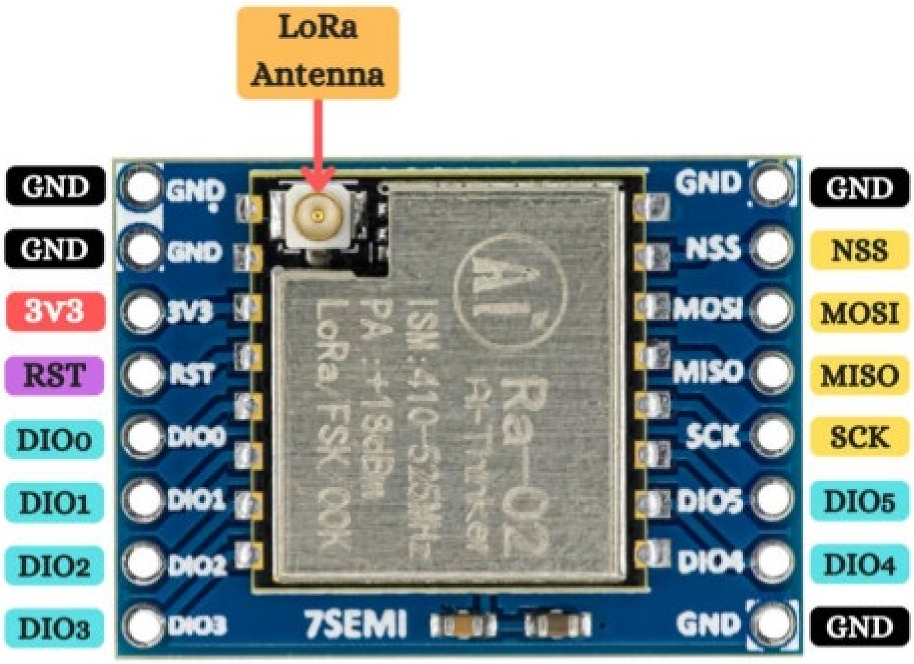
Normal timing expectations

1. **Scheduled LoRa transmit**: about once per minute (per vest).
2. **Event LoRa transmit**: immediately when the sensor transitions from dry to wet.

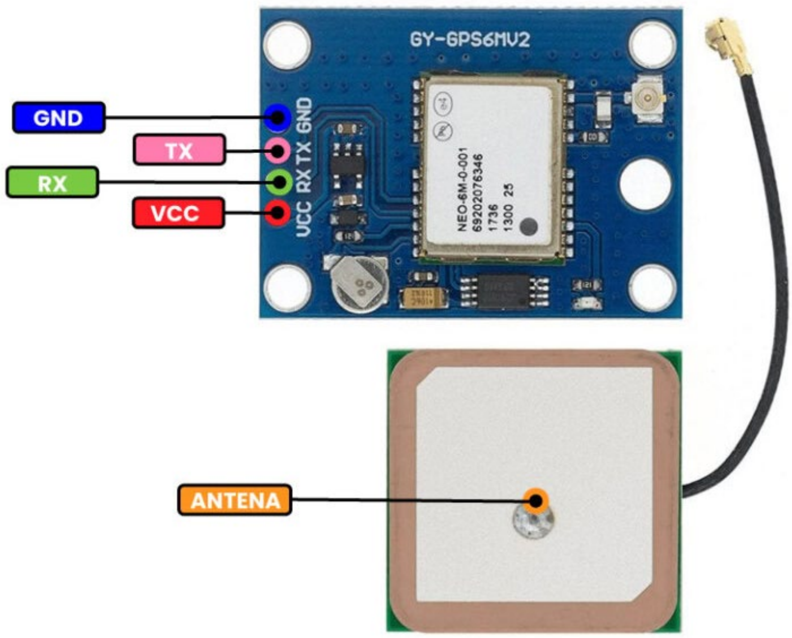
A7. ESP32 Pin Out



A7. LoRa SX1278 Pin Out



A8. GY-NEO6MV2 Pin Out



A9. Rain Sensor Pin Out